In [2]:

```
# import library
from google.colab import files
uploaded = files.upload()
import pandas as pd
import matplotlib.pyplot as plt
```

Choose files No file chosen Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

```
Saving sales data-set.csv to sales data-set.csv
```

In [3]:

```
# load the data set
data = pd.read_csv("sales data-set.csv")
print(data.head())
```

```
   Store  Dept        Date  Weekly_Sales  IsHoliday
0      1     1  05/02/2010      24924.50      False
1      1     1  12/02/2010      46039.49       True
2      1     1  19/02/2010      41595.55      False
3      1     1  26/02/2010      19403.54      False
4      1     1  05/03/2010      21827.90      False
```

In [4]:

```
# get information of data set
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 421570 entries, 0 to 421569
Data columns (total 5 columns):
 #   Column        Non-Null Count   Dtype
---  ------        --------------   -----
 0   Store         421570 non-null  int64
 1   Dept          421570 non-null  int64
 2   Date          421570 non-null  object
 3   Weekly_Sales  421570 non-null  float64
 4   IsHoliday     421570 non-null  bool
dtypes: bool(1), float64(1), int64(2), object(1)
memory usage: 13.3+ MB
```

In [5]:

```
# get the shape of data, number of row and column
data.shape
```

Out [5]: (421570, 5)

In [6]:

```
# get the null values
data.isnull().sum()
```

Out [6]:

	0
Store	0

	0
Dept	0
Date	0
Weekly_Sales	0
IsHoliday	0

dtype: int64

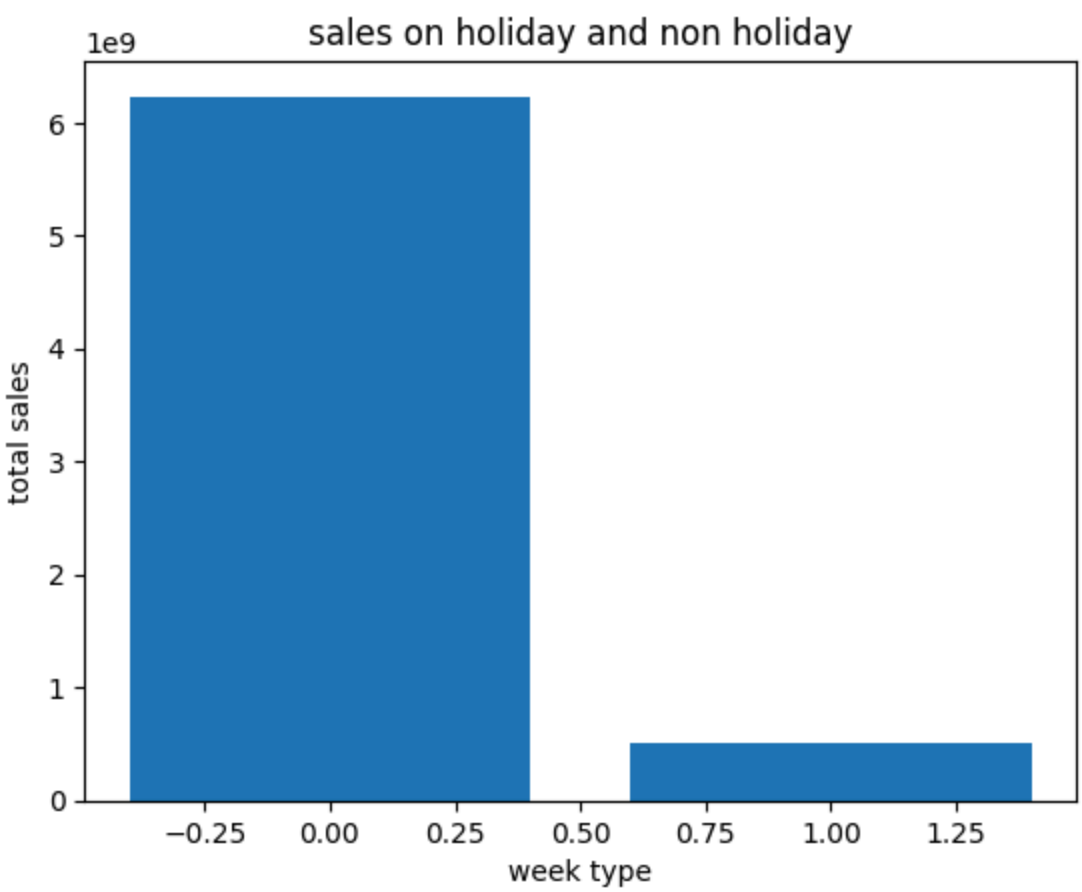
In [8]:

```
# The day with the higest sales
Isholiday_sales = data.groupby(data["IsHoliday"]).sum()["Weekly_Sales"].sort_v
print(Isholiday_sales)
```

```
IsHoliday
False    6.231919e+09
True     5.052996e+08
Name: Weekly_Sales, dtype: float64
```

In [11]:

```
# visualization of the sales
plt.bar(Isholiday_sales.index, Isholiday_sales.values)
plt.ylabel('total sales')
plt.xlabel('week type')
plt.title('sales on holiday and non holiday')
plt.show()
```



In [13]:

```
holiday_data = data[data["IsHoliday"] == True]

print(holiday_data)
```

```
        Store  Dept        Date  Weekly_Sales  IsHoliday
1           1     1  12/02/2010      46039.49       True
31          1     1  10/09/2010      18194.74       True
42          1     1  26/11/2010      18820.29       True
47          1     1  31/12/2010      19124.58       True
53          1     1  11/02/2011      37887.17       True
...       ...   ...         ...           ...        ...
421510     45    98  09/09/2011       1092.66       True
421521     45    98  25/11/2011        794.62       True
421526     45    98  30/12/2011        553.21       True
421532     45    98  10/02/2012        770.86       True
421562     45    98  07/09/2012        352.44       True

[29661 rows x 5 columns]
```

In [15]:

```
#total holiday sales
holiday_sales = data[data["IsHoliday"] == True]["Weekly_Sales"].sum()

print(holiday_sales)
```

```
505299551.56
```

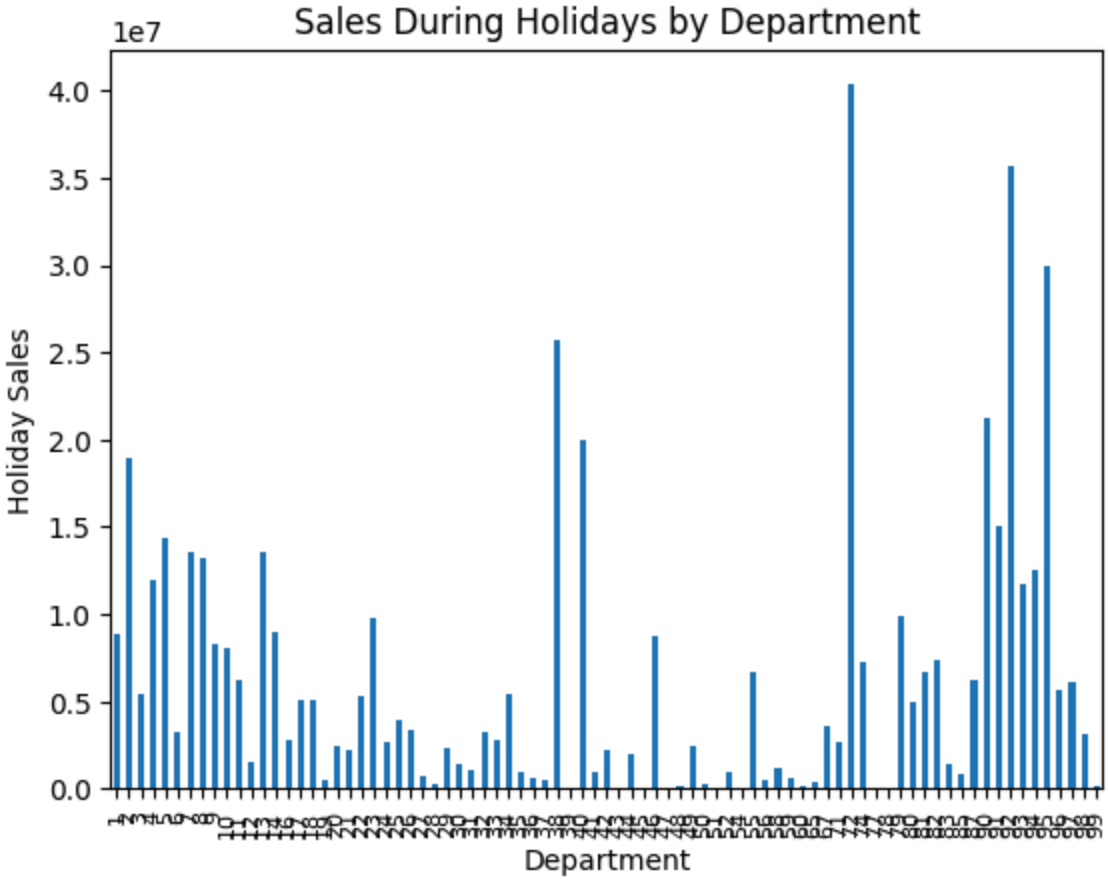
In [16]:

```
# holiday sales by department
holiday_sales_dept = data[data["IsHoliday"] == True].groupby("Dept")["Weekly_S

print(holiday_sales_dept)
```

```
Dept
1      8867168.77
2     18991468.92
3      5481134.64
4     12005820.97
5     14347362.90
         ...
95    29996233.40
96     5723180.51
97     6133824.50
98     3191644.19
99      119272.58
Name: Weekly_Sales, Length: 81, dtype: float64
```

In [17]:

```
#Visualization
holiday_sales_dept.plot(kind="bar")
plt.xlabel("Department")
plt.ylabel("Holiday Sales")
plt.xticks(rotation='vertical', size=8)
plt.title("Sales During Holidays by Department")
plt.show()
```



In [18]:

```
# department with high holiday sales
top_holiday_sales_by_dept=holiday_sales_dept.sort_values(ascending=False)
```

In [19]:

```
top_holiday_sales_by_dept
```

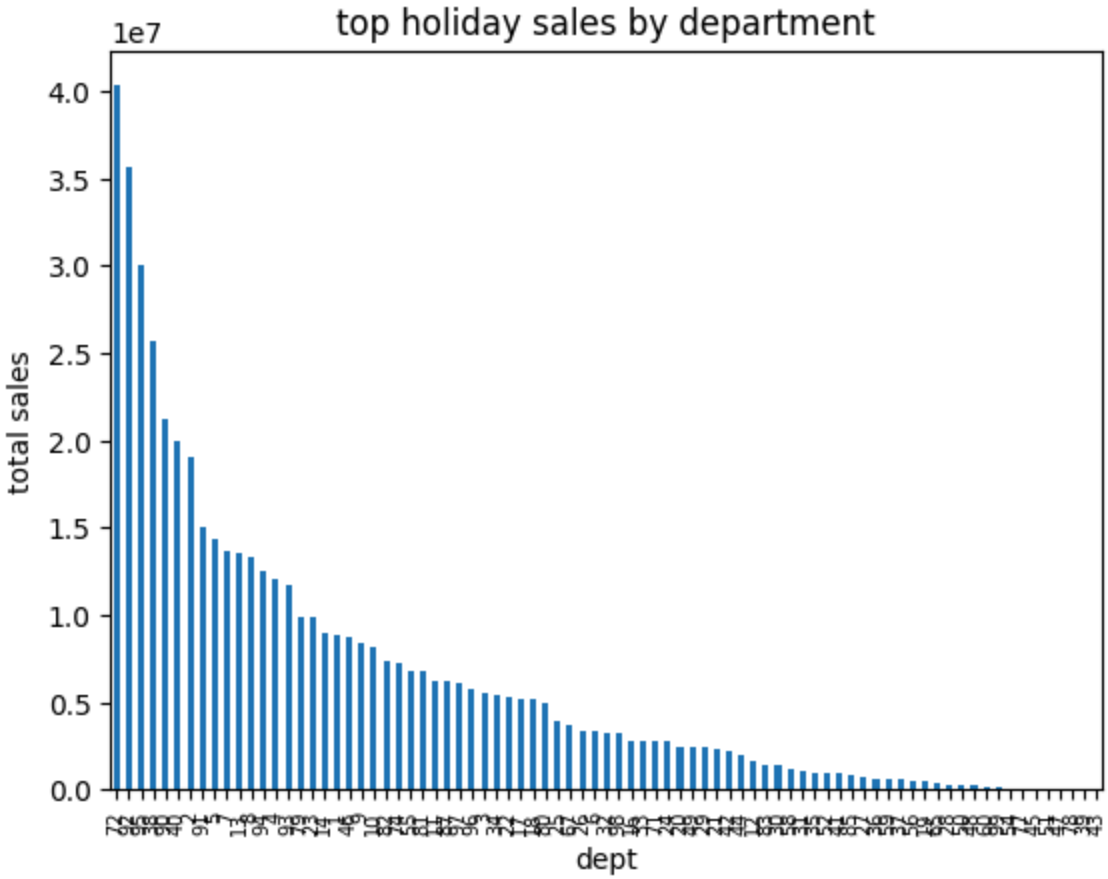
Out [19]:

Dept	Weekly_Sales
72	40313605.46
92	35695397.10
95	29996233.40
38	25682245.57
90	21209128.22
...	...
51	1711.58
47	250.93
78	169.76
39	13.41
43	0.25

81 rows × 1 columns

dtype: float64

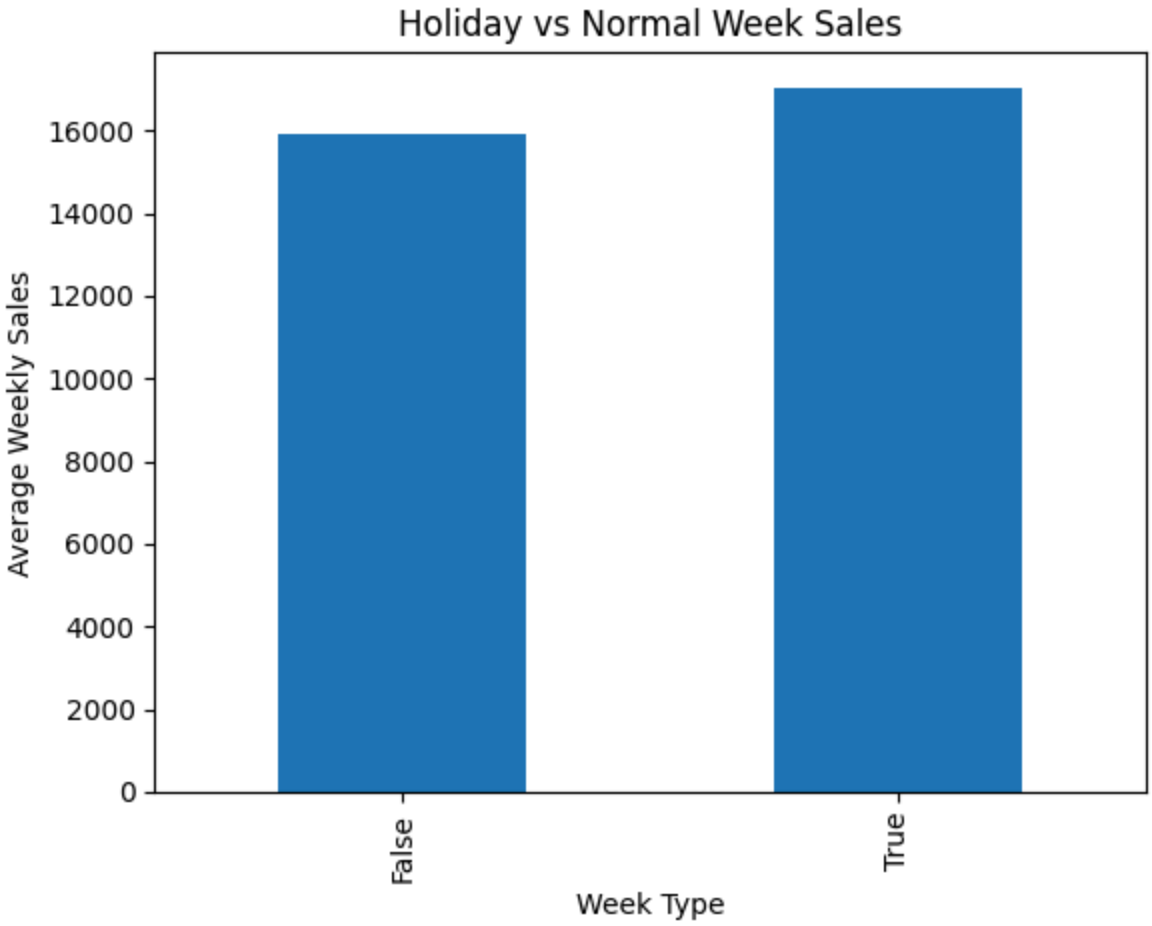
In [24]:
```
top_holiday_sales_by_dept.plot(kind="bar")
plt.xlabel('dept')
plt.ylabel('total sales')
plt.title('top holiday sales by department')
plt.xticks(rotation='vertical', size=7)
plt.show()
```



In [28]:
```
 #Comparison btn hiliday week sales and non holiday week sales
sales_compare = data.groupby("IsHoliday")["Weekly_Sales"].mean()

print(sales_compare)
```

```
IsHoliday
False    15901.445069
True     17035.823187
Name: Weekly_Sales, dtype: float64
```

In [29]:
```
#visualization of the holiday and normal week sales
sales_compare.plot(kind="bar")
plt.xlabel("Week Type")
plt.ylabel("Average Weekly Sales")
plt.title("Holiday vs Normal Week Sales")
plt.show()
```



In [33]:

```
# convert the date inform of month_year
data["month_year"] = pd.to_datetime(data["Date"], format="%d/%m/%Y").dt.strfti
data["month_year"]
```

Out [33]:

	month_year
0	10-02
1	10-02
2	10-02
3	10-02
4	10-03
...	...
421565	12-09
421566	12-10
421567	12-10
421568	12-10
421569	12-10

421570 rows × 1 columns

dtype: object

In [34]:

```
# the month with the highest sales
data.groupby(["month_year"]).sum()["Weekly_Sales"].sort_values(ascending=False
```

Out [34]:

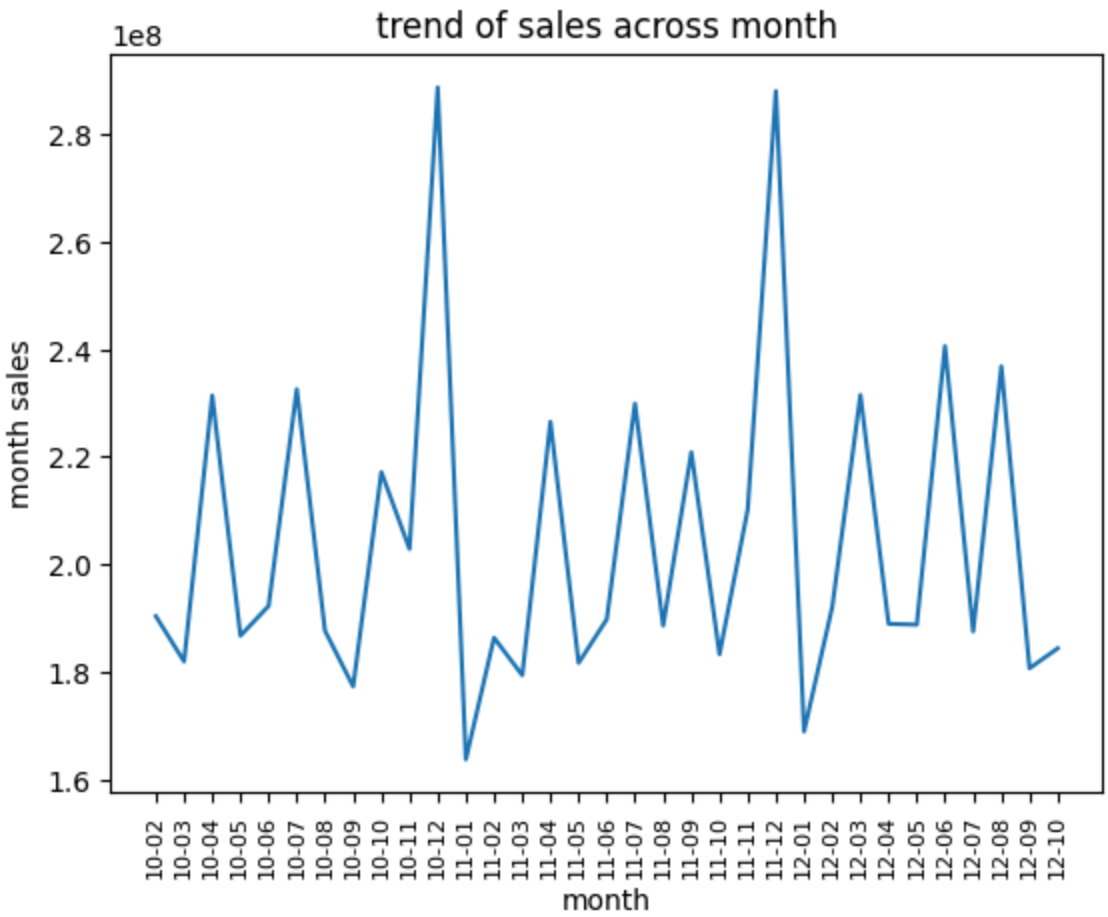
month_year	Weekly_Sales
10-12	2.887605e+08
11-12	2.880781e+08
12-06	2.406103e+08
12-08	2.368508e+08
10-07	2.325801e+08
12-03	2.315097e+08
10-04	2.314124e+08
11-07	2.299114e+08
11-04	2.265265e+08
11-09	2.208477e+08
10-10	2.171618e+08
11-11	2.101624e+08
10-11	2.028534e+08
10-06	1.922462e+08
12-02	1.920636e+08
10-02	1.903330e+08
11-06	1.897734e+08
12-04	1.889209e+08
12-05	1.887665e+08
11-08	1.885993e+08
10-08	1.876401e+08
12-07	1.875095e+08
10-05	1.867109e+08
11-02	1.863313e+08
12-10	1.843617e+08
11-10	1.832613e+08
10-03	1.819198e+08
11-05	1.816482e+08
12-09	1.806455e+08
11-03	1.793564e+08
10-09	1.772679e+08

month_year	Weekly_Sales
12-01	1.688945e+08
11-01	1.637040e+08

dtype: float64

In [35]:

```
# the trend of sales
sales_trend = data.groupby(["month_year"]).sum()["Weekly_Sales"]
plt.plot(sales_trend.index, sales_trend.values)
plt.ylabel('month sales')
plt.xlabel('month')
plt.title('trend of sales across month')
plt.xticks(rotation='vertical', size=8)
plt.show()
```



In [13]:

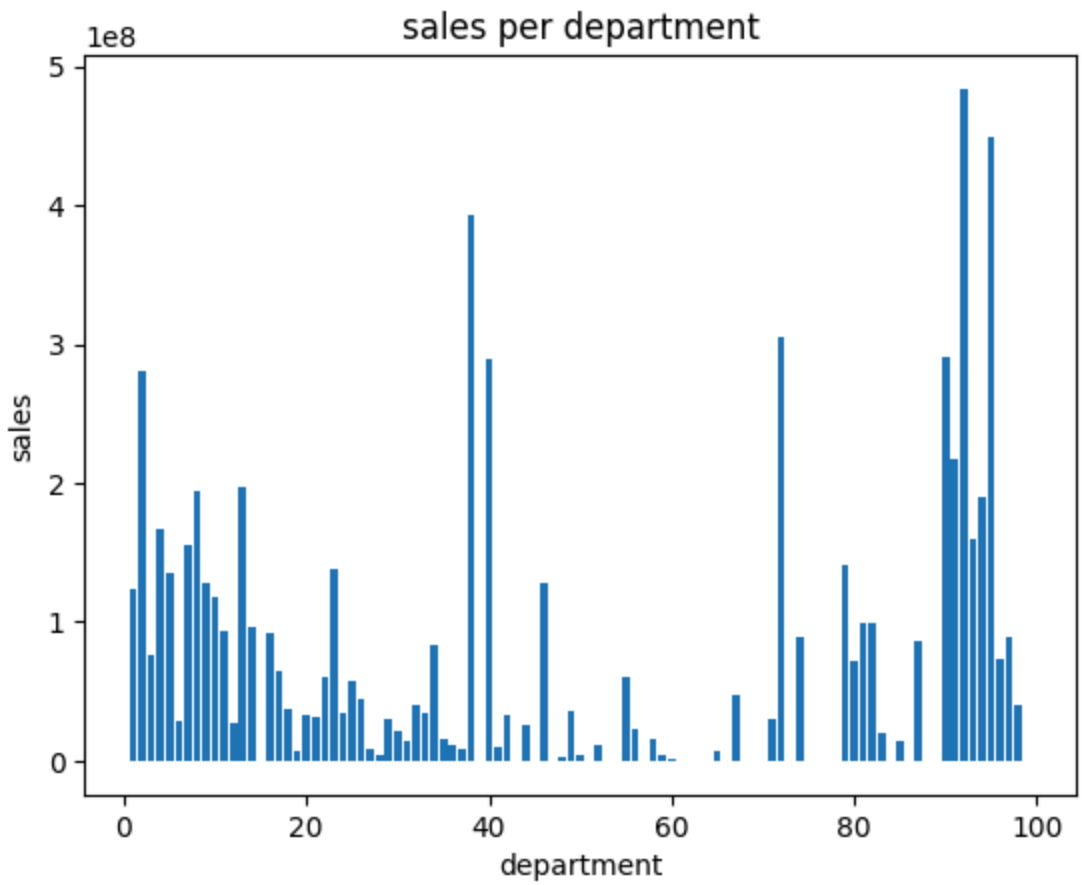
```
mostsold_dept=data.groupby(["Dept"]).sum()["Weekly_Sales"].sort_values(ascendi
print(mostsold_dept)
```

```
Dept
92    4.839433e+08
95    4.493202e+08
38    3.931181e+08
72    3.057252e+08
90    2.910685e+08
          ...
51    3.057283e+04
```

```
78    1.714710e+03
39    1.779800e+02
43    1.432000e+01
47   -4.962930e+03
Name: Weekly_Sales, Length: 81, dtype: float64
```

In [17]:

```
plt.bar(mostsold_dept.index, mostsold_dept.values)
plt.ylabel('sales')
plt.xlabel('department')
plt.title('sales per department')
plt.show()
```



In [33]:

```
data["Store"].max()
```

Out [33]: 45

In [34]:

```
data["Store"].min()
```

Out [34]: 1

In [36]:

```
data["Store"]
```

Out [36]:

	Store
0	1
1	1
2	1
3	1

	Store
4	1
...	...
421565	45
421566	45
421567	45
421568	45
421569	45

421570 rows × 1 columns

dtype: int64

In [41]:

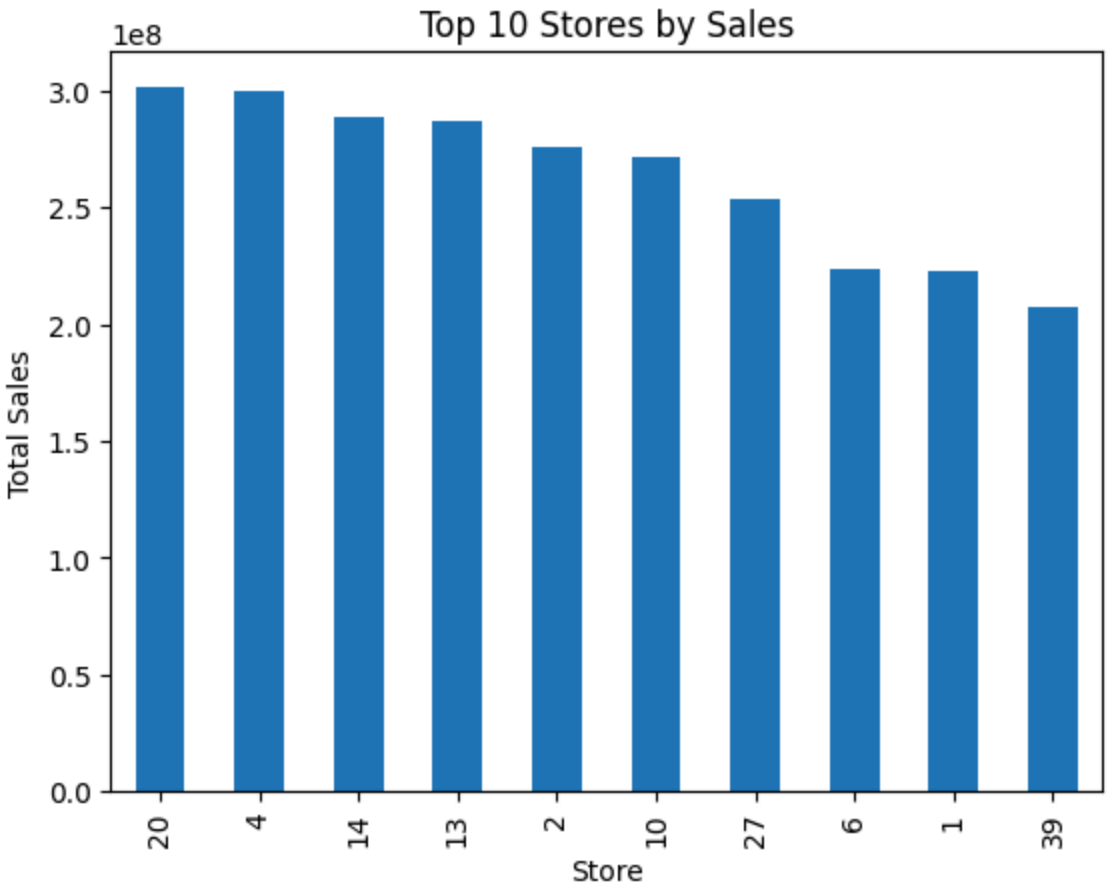
```
# top performing store by sales
top_store=data.groupby(["Store"]).sum()["Weekly_Sales"].sort_values(ascending=
print(top_store)
```

```
Store
20    3.013978e+08
4     2.995440e+08
14    2.889999e+08
13    2.865177e+08
2     2.753824e+08
10    2.716177e+08
27    2.538559e+08
6     2.237561e+08
1     2.224028e+08
39    2.074455e+08
19    2.066349e+08
31    1.996139e+08
23    1.987506e+08
24    1.940160e+08
11    1.939628e+08
28    1.892637e+08
41    1.813419e+08
32    1.668192e+08
18    1.551147e+08
22    1.470756e+08
12    1.442872e+08
26    1.434164e+08
34    1.382498e+08
40    1.378703e+08
35    1.315207e+08
8     1.299512e+08
17    1.277821e+08
45    1.123953e+08
21    1.081179e+08
25    1.010612e+08
43    9.056544e+07
15    8.913368e+07
7     8.159828e+07
42    7.956575e+07
9     7.778922e+07
29    7.714155e+07
16    7.425243e+07
37    7.420274e+07
30    6.271689e+07
3     5.758674e+07
38    5.515963e+07
36    5.341221e+07
5     4.547569e+07
44    4.329309e+07
```

```
33    3.716022e+07
Name: Weekly_Sales, dtype: float64
```

In [43]:

```
#visualization of the top 10
top_store.head(10).plot(kind="bar")
plt.xlabel("Store")
plt.ylabel("Total Sales")
plt.title("Top 10 Stores by Sales")
plt.show()
```



In [48]:

```
#Store with high holiday sales
holiday_store_sales = data[data["IsHoliday"]==True].groupby("Store")["Weekly_S

print(holiday_store_sales.sort_values(ascending=False))
```

```
Store
20    22490350.81
4     22431026.24
14    21205829.98
10    21137559.49
13    21130438.06
2     20792669.00
27    18922992.78
6     16809079.27
1     16657476.56
19    15770467.34
39    15511274.80
31    15000260.30
28    14782446.05
24    14750982.51
23    14625422.94
11    14483944.85
```

```
41    13349478.56
32    12037840.83
18    11694221.61
12    11381404.20
22    10848746.56
35    10743484.57
26    10720468.49
34    10419780.89
40    10080340.75
17     9797969.71
8      9753308.60
45     8362937.13
21     8264913.09
25     7396768.42
15     7064060.18
7      6724002.65
43     6359462.78
29     6069578.89
9      5889508.21
42     5676941.58
16     5667336.46
37     5075250.50
3      4378110.50
30     4368593.07
38     3815098.78
36     3676406.30
5      3595016.07
44     2960356.01
33     2625945.19
Name: Weekly_Sales, dtype: float64
```

In [49]:

```
holiday_store_sales.head(10).plot(kind="bar")
plt.xlabel("Store")
plt.ylabel("Total holiday store by Sales")
plt.title("Top 10 holiday Stores by Sales")
plt.show()
```

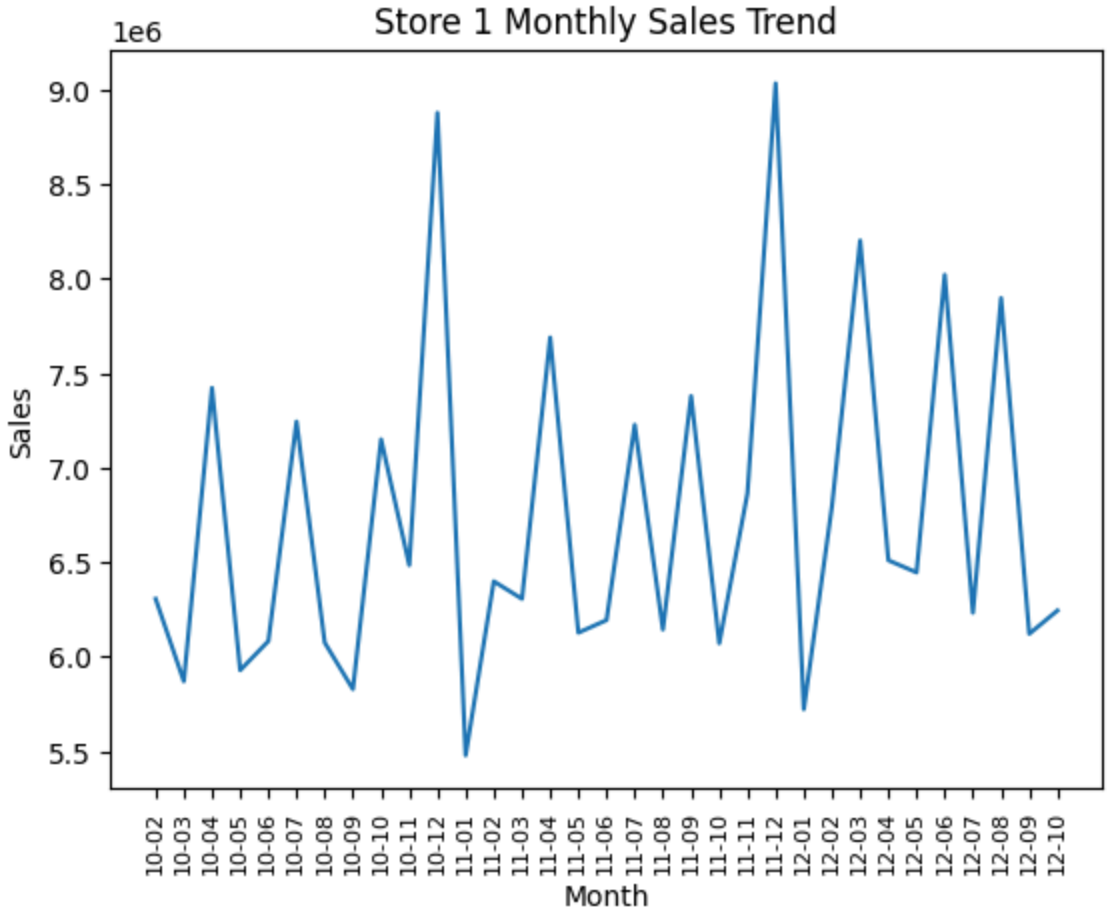
In [50]:

```
#monthly sales for each store
store_trend = data.groupby(["month_year","Store"])["Weekly_Sales"].sum()

print(store_trend)
```

```
month_year  Store
10-02       1        6307344.10
            2        8264347.77
            3        1711198.23
            4        8299040.36
            5        1202728.00
                        ...
12-10       41       5452445.75
            42       2261705.49
            43       2473507.39
            44       1360020.41
            45       2946326.39
Name: Weekly_Sales, Length: 1485, dtype: float64
```

In [52]:

```
# trend in monthly sales in store 1
store1 = data[data["Store"] == 1].groupby("month_year")["Weekly_Sales"].sum()

plt.plot(store1.index, store1.values)
plt.xlabel("Month")
plt.ylabel("Sales")
plt.xticks(rotation='vertical', size=8)
plt.title("Store 1 Monthly Sales Trend")
plt.show()
```



In [27]:

```
# department in store 20
store20 = data[data["Store"] == 20]
print(store20)
```

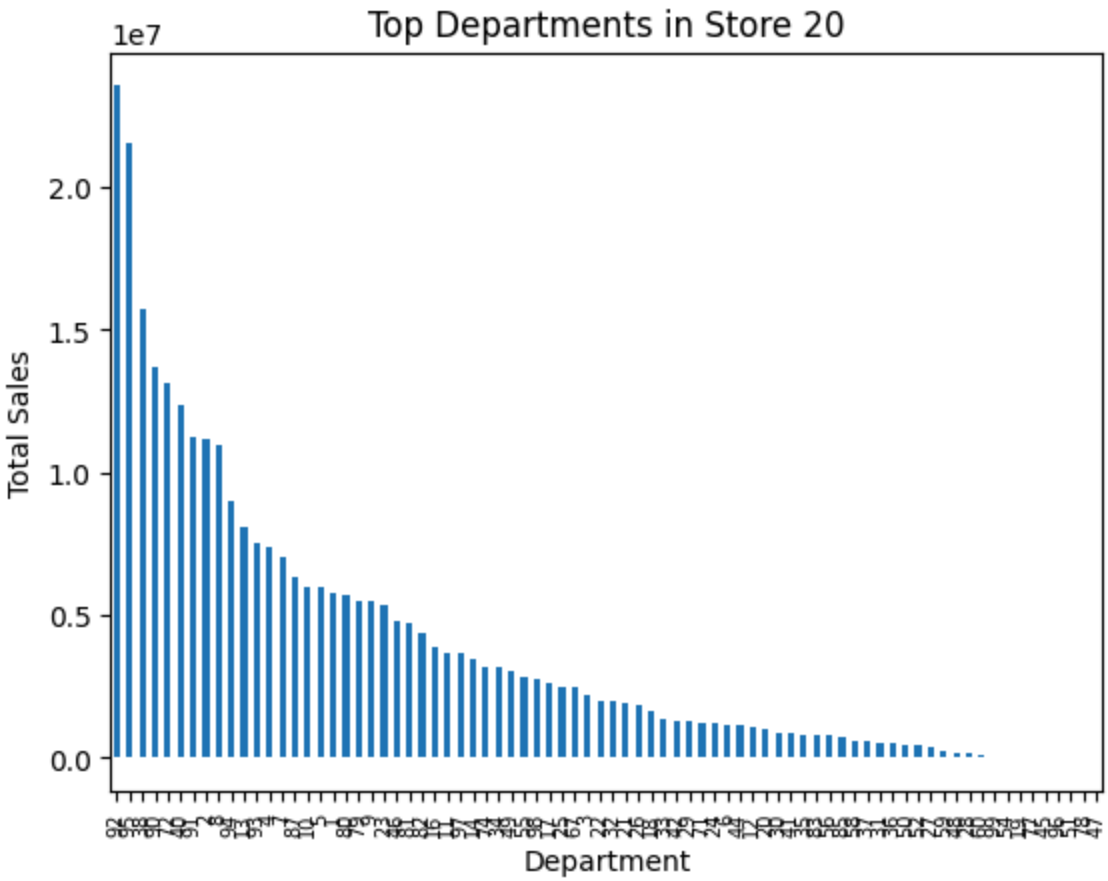
```
        Store  Dept        Date  Weekly_Sales  IsHoliday
187335     20     1  05/02/2010      46021.21      False
187336     20     1  12/02/2010      52734.17       True
187337     20     1  19/02/2010      55649.79      False
187338     20     1  26/02/2010      36805.42      False
187339     20     1  05/03/2010      41206.78      False
...       ...   ...         ...           ...        ...
197544     20    99  07/09/2012          0.67       True
197545     20    99  14/09/2012          0.05      False
197546     20    99  21/09/2012         29.91      False
197547     20    99  05/10/2012        660.00      False
197548     20    99  12/10/2012        200.00      False

[10214 rows x 5 columns]
```

In [28]:

```
dept_sales = store20.groupby("Dept")["Weekly_Sales"].sum().sort_values(ascendi
print(dept_sales)
```

```
Dept
92    23542625.04
95    21537795.62
38    15759345.93
90    13707777.99
72    13154550.44
         ...
45         666.18
96         244.27
51         161.18
78          29.00
```

```
47        -378.58
Name: Weekly_Sales, Length: 78, dtype: float64
```

In [30]:

```
# department in store 20 wich generate high sales
dept_sales.plot(kind="bar")
plt.title("Top Departments in Store 20")
plt.xlabel("Department")
plt.ylabel("Total Sales")
plt.xticks(size=7)
plt.show()
```